

[Dashboard](#)[my courses](#)[AMCN SS 2022](#)[wireless transmission](#)[Quiz: Wireless Transmission - Cellular Networks \(Capacity and Interference\)](#)

Started at	Sunday, April 17, 2022 at 1:13 p.m
status	Completed
Finished at	Sunday, April 17, 2022, 1:17 p.m
Spent time	3 minutes 53 seconds
valuation	4.00 out of 4.00 (100 %)
feedback	Congratulations, excellent performance (90-100%).

question **1**

Complete

Score 1.00 out of 1.00

Compare the spatial reuse of 3 with a reuse of 9 with respect to capacity (data rate) and quality (error rate). Assume the same total frequency spectrum and same modulation and coding schemes for both schemes.

Choose one or more answers:

- ☐ a. Reuse 3 provides lower raw throughput rate but with higher quality, ie lower error rates
- ☒ b. Reuse 3 provides higher raw throughput rate but with higher error rates

question **2**

Complete

Score 1.00 out of 1.00

How can the capacity of a cellular system be improved assuming that we have an interference-limited system?

Choose one or more answers:

- ☒ a. Use better receivers that can tolerate a lower signal-to-interference ratio
- ☐ b. Increase the transmit power of the base stations and the users
- ☒ c. Increase the spatial diversity using multi-sector antennas
- ☒ i.e. Prefer users at the cell center instead of users at the cell edge
- ☒ e. Increase the number of base stations

Frage 3

Vollständig

Erreichte
Punkte 1,00 von
1,00

Under which circumstances is direct communication between two mobiles a solution to improve the network capacity of cellular networks?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. When the two mobiles are close to each other and reside in the same cell
- ☐ b. When the two mobiles are close to the cell edge but far from each other
- ☐ c. When the two mobiles are far from each other and reside in different cells
- ☒ d. When the two mobiles are close to each other and reside in different cells

Frage 4

Vollständig

Erreichte
Punkte 1,00 von
1,00

What are the results of adding more base stations to a given cellular network layout assuming that there is no extra spectrum available?
What are the additional measures that need to be taken into account to handle this?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Adapt the transmit power of base stations
- ☒ b. Adapt the transmit power of mobiles

Vorhergehende Aktivität

◀ Quiz: Wireless
Transmission - Modulation
and Coding

Direkt zu:



next activity

Medium access schemes ►



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